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Surgical, Radiographic, and Endoscopic Anatomy of the Male Reproductive System

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Keywords

anatomy; ejaculatory duct; epididymis; penis; prostate; scrotum; seminal vesicle; testicle; urethra; vas deferens

Questions

1. The route that sperm travel through from the seminiferous tubules to the epididymis, in order, is:
    a. straight tubules, efferent ductules, rete testis.
    b. rete testis, straight tubules, efferent ductules.
    c. efferent ductules, rete testis, straight tubules.
    d. straight tubules, rete testis, efferent ductules.
    e. rete testis, efferent ductules, straight tubules.
2. The testis is enveloped by a tough capsule composed from external to internal, in order, of the:
    a. visceral tunica vaginalis, tunica albuginea, tunica vasculosa.
    b. tunica vasculosa, tunica albuginea, visceral tunica vaginalis.
    c. tunica albuginea, tunica vasculosa, visceral tunica vaginalis.
    d. visceral tunica vaginalis, tunica vasculosa, tunica albuginea.
    e. tunica albuginea, visceral tunica vaginalis, tunica vasculosa.

3. What percentage of testicular volume is made up by interstitium?
    a. 10% to 20%
    b. 20% to 30%
    c. 30% to 40%
    d. 40% to 50%
    e. 50% to 60%
4. Which of the following is the main arterial supply to the testis?
    a. Testicular artery
    b. Superficial dorsal artery
    c. The artery of the vas deferens
    d. Cremasteric artery
    e. Deep dorsal artery
5. The lymphatic drainage from the testis drains to:
    a. superficial and deep inguinal nodes.
    b. external and internal iliac nodes.
    c. para-aortic and interaortocaval nodes.
    d. internal iliac and obturator nodes.
    e. external iliac and obturator nodes.
6. The cremaster muscle is innervated by the:
    a. ilioinguinal nerve.
    b. genital branch of the genitofemoral nerve.
    c. femoral branch of the genitofemoral nerve.
    d. terminal branches of the subcostal nerve (T12).
    e. iliohypogastric nerve.
7. The nerves thought to be the main contributors to pain in men with chronic orchialgia include:
    a. perivasal complex.
    b. posterior periarterial/lipomatous complex.
    c. intracremasteric complex.
    d. femoral branch of the genitofemoral nerve.
    e. a, b, and c.
8. The anatomic component of the blood-testis barrier is the:
    a. tight junctions between Sertoli cells.
    b. myoendothelial junctions.
    c. tunica albuginea.
    d. basement membrane of seminiferous tubules.

e. centrifugal arteries.

9. The ducts of which of the following prostatic zones drain into the preprostatic urethra?
    a. Periurethral glands
    b. Central zone
    c. Transition zone
    d. Peripheral zone
    e. a and c
10. Benign prostatic hyperplasia (BPH) arises from the:
    a. periurethral glands.
    b. central zone.
    c. transition zone.
    d. peripheral zone.
    e. a and c.
11. Lymphatic drainage from the prostate drains to the:
    a. external iliac and common iliac nodes.
    b. internal iliac and obturator nodes.
    c. para-aortic nodes.
    d. internal iliac and inguinal nodes.
    e. perirectal and common iliac nodes.
12. The apex of the prostate is continuous with the:
    a. bladder neck.
    b. pubococcygeal portion of the of the levator ani.
    c. arcus tendineus fascia pelvis.
    d. bulbar urethra.
    e. striated urethral sphincter.
13. Lymphatic drainage from the bulbar urethra travels:
    a. through the perianal nodes to reach the pelvis.
    b. directly to the deep pelvic lymph nodes.
    c. through the superficial and deep inguinal lymph nodes.
    d. to the prepubic nodes.
    e. to para-aortic lymph nodes along with testicular drainage.
14. The only segment of the urethra that does not have transitional epithelium is the:
    a. prostatic urethra.
    b. membranous urethra.

c. bulbar urethra.
d. bulbomembranous urethra.
e. fossa navicularis.

15. The glans penis is the most distal expansion of the:
    a. corpus cavernosum.
    b. corpus spongiosum.
    c. prepuce.
    d. urethra.
    e. Buck's fascia.
16. The dartos layer of smooth muscle and fascia in the scrotum is continuous with:
    a. the dartos layer of the penis.
    b. Colles fascia.
    c. Scarpa fascia.
    d. Buck's fascia.
    e. a, b, and c.

Answers

1. **d. Straight tubules, rete testis, efferent ductules.** Septa form 200 to 300 cone-shaped lobules, each containing one or more convoluted seminiferous tubules. Each tubule is U-shaped and has a stretched length of nearly 1 m. Interstitial (Leydig) cells lie in the loose tissue surrounding the tubules and are responsible for testosterone production. Toward the apices of the lobules, the seminiferous tubules become straight (tubuli recti) and enter the mediastinum testis to form an anastomosing network of tubules lined by flattened epithelium. **This network, known as the *rete testis*, forms 12 to 20 efferent ductules and passes into the largest portion of epididymis, the caput.**
2. **a. Visceral tunica vaginalis, tunica albuginea, tunica vasculosa.** The testis is enveloped by a tough capsule composed, from external to internal, of the visceral tunica vaginalis, the tunica albuginea, and the tunica vasculosa, before reaching the parenchyma of the

testis. **The tunica albuginea is composed of smooth muscle cells that pass through collagenous tissue.**

3. **b. 20% to 30%. The testicular interstitial tissue includes Leydig cells, mast cells, macrophages, nerves, blood vessels, and lymphatic vessels.** This interstitial tissue makes up 20% to 30% of the testicular volume.
4. **a. Testicular artery. There are three arterial supplies to the testis: the testicular (internal spermatic) artery, the artery of the vas deferens (deferential artery), and the cremasteric (external spermatic) artery. The testicular artery is the main blood supply to the testis, and its diameter is greater than the deferential and cremasteric arteries combined.**
5. **c. Para-aortic and interaortocaval nodes.** Lymphatic channels from the testis drain into the para-aortic and interaortocaval lymph nodes. These lymphatic channels ascend within the spermatic cord after leaving the testis.
6. **b. Genital branch of the genitofemoral nerve.** The genital branch of the genitofemoral nerve follows the spermatic cord through the inguinal canal, supplies the cremaster muscle, and supplies sensation to the anterior scrotum.
7. **e. a, b, and c.** Three distinct anatomic distributions of nerves have been isolated within the spermatic cord and are thought to be the primary contributors in men with chronic orchialgia. These include a perivasal complex, posterior periarterial/lipomatous complex, and intracremasteric complex.
8. **a. Tight junctions between Sertoli cells. There are extremely strong tight junctions between Sertoli cells, which provide an intracellular barrier that allows for spermatogenesis in an immune-privileged site.** This is the barrier known as the *blood-testis barrier.*
9. **a. Periurethral glands.** At its midpoint, the urethra turns approximately 35 degrees anteriorly, but this angulation can vary from 0 to 90 degrees. This angle divides the prostatic urethra into proximal (preprostatic) and distal (prostatic) segments, which are functionally and anatomically discrete. Small periurethral glands, lacking periglandular smooth muscle, extend between the fibers of

the longitudinal smooth muscle to be enclosed by the preprostatic sphincter.

10. **e. a and c.** The periurethral glands can contribute significantly to prostatic volume in older men as one of the sites of origin of BPH. The transition zone commonly gives rise to BPH.
11. **b. Internal iliac and obturator nodes.** Lymphatic drainage is primarily to the obturator and internal iliac nodes.
12. **e. Striated urethral sphincter.** The prostate is enveloped by a collagen, elastin, and smooth muscle capsule. The capsule measures 0.5 mm in thickness posteriorly and laterally on average. **There is no true prostatic capsule at the apex of the prostate, where normal prostate glands are seen blending into the striated muscle of the urethral sphincter.**
13. **c. Through the superficial and deep inguinal lymph nodes.** The penis, scrotum, and perineum drain into the inguinal lymph nodes. These nodes can be divided into superficial groups and deep groups.
14. **e. Fossa navicularis. Unlike the transitional epithelium of the remainder of the urethra, the urethral mucosa that traverses the glans penis is squamous epithelium.** These cells become keratinized near the meatus.
15. **b. Corpus spongiosum.** The glans penis is an expansion of the corpus spongiosum.
16. **e. a, b, and c.** The dartos layer of smooth muscle is anatomically continuous with Colles fascia, Scarpa fascia, and the dartos fascia of the penis.

Chapter Review

1. The normal testicle measures 2.5 × 3 × 4 cm and has a volume of 15 to 25 mL.
2. Both ends of the seminiferous tubules end in the rete testis.
3. Although there are three arteries that supply the testis, the testicular artery is the most important, and its ligation may result in testicular atrophy.

4. The testis, epididymis, vas, and seminal vesicles are innervated by the autonomic system, which is made up of afferent and efferent nerves.
5. The anatomic zones of the prostate include the transition zone, which is the site of benign prostatic hyperplasia; the central zone, which contains the ejaculatory ducts (glands arising in this zone are thought to be derived from the wolffian duct, unlike the glands in the rest of the prostate); the peripheral zone, which contains 70% of the glandular epithelium and where 70% of prostate cancers occur; and the anterior zone, which is made up of fibromuscular stroma.
6. Seventy percent of the prostate is made up of glandular tissue, and 30% is fibromuscular tissue.
7. A permeable septum separates the corpora cavernosa, one from the other, and allows for free communication between the vascular spaces of the corpora bodies.
8. Scrotal lymphatics do not cross the midline and drain into the ipsilateral groin, unlike the penis, where the lymphatics cross over, allowing drainage to either groin irrespective of the side of the lesion.
9. The rete testis forms 12 to 20 efferent ductules and passes into the largest portion of epididymis, the caput.
10. The tunica albuginea is composed of smooth muscle cells that pass through collagenous tissue.
11. The genital branch of the genitofemoral nerve follows the spermatic cord through the inguinal canal, supplies the cremaster muscle, and supplies sensation to the anterior scrotum.
12. There are extremely strong tight junctions between Sertoli cells, which provide an intracellular barrier that allows for spermatogenesis in an immune-privileged site.
13. There is no true prostatic capsule at the apex of the prostate.
14. The penis, scrotum, and perineum drain into the inguinal lymph nodes.
15. The urethral mucosa that traverses the glans penis is squamous epithelium.

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Male Reproductive Physiology

Paul J. Turek

Keywords

azoospermia; ejaculatory duct; epididymis; epigenetics; obstructive azoospermia; prolactin; spermatogenesis; spermatogonial stem cells; spermatozoa; testicle; testicular failure; testosterone; vas deferens

Questions

1. Embryologically, the vas deferens and body of the epididymis are derived from what developmental structure?
    a. Müllerian ducts
    b. Wolffian ducts
    c. Urogenital ridge
    d. Gubernaculum testis
    e. Metanephros
2. The proportion of testicular volume comprised of seminiferous tubules in the human testis is:
    a. 5%
    b. 10% to 20%
    c. 50% to 60%
    d. 70% to 80%
    e. 90% to 95%

3. Which hormones play a central role in regulation of Sertoli cell function?
    a. LH, follicle-stimulating hormone (FSH)
    b. FSH, estradiol
    c. Prolactin, LH
    d. FSH, testosterone
    e. Androgen-binding protein (ABP), testosterone
4. The vast majority of the fluid in the male ejaculate is derived from the:
    a. epididymides
    b. ejaculatory ducts
    c. seminal vesicles
    d. testicles
    e. vas deferens
5. Testicular blood supply is derived from all of the following sources EXCEPT:
    a. Internal spermatic artery
    b. Deferential artery
    c. External spermatic artery
    d. Cremasteric artery
    e. Pudendal artery
6. A 25-year old bodybuilder eschews the merits of "natural" bodybuilding and uses injectable anabolic steroids regularly to achieve muscle bulk. While on anabolic steroids, his fertility potential would be expected to be:
    a. normal, because exogenous testosterone does not impair production of endogenous testosterone.
    b. low, because exogenous testosterone stimulates pituitary production of FSH and LH
    c. low, because exogenous testosterone inhibits pituitary production of FSH and LH
    d. low, because exogenous testosterone is not as potent as endogenous testosterone at nurturing spermatogenesis.
    e. normal, because intratesticular testosterone concentrations are 50 times higher than serum levels, whether or not the blood contains exogenous testosterone.

7. Structural elements of the blood-testis barrier are observed in:
    a. Sertoli-Sertoli cell junctions
    b. Leydig cells
    c. basement membranes
    d. desmosomes between spermatocytes
    e. spermatids
8. How do elevated levels of prolactin influence testosterone production?
    a. Inhibit gonadotropin-releasing hormone (GnRH) and LH
    b. Indirectly inhibit Sertoli cells
    c. Directly inhibit Leydig cells
    d. Upregulate inhibin
    e. Downregulate activin
9. What is the normal developmental pattern of spermatogenic cells within the seminiferous tubule?
    a. Sertoli, spermatogonia, spermatocyte
    b. Spermatocyte, spermatogonia, spermatid
    c. Spermatid, Sertoli, spermatocyte
    d. Spermatogonia, spermatid, spermatocyte
    e. Spermatogonia, spermatocyte, spermatid
10. What testicular hormone is the major feedback inhibitor of LH secretion?
    a. Testosterone
    b. Inhibin
    c. Activin
    d. Prolactin
    e. Sertolin
11. What is the normal size for adult human testes?
    a. 5 mL
    b. 10 mL
    c. 20 mL
    d. 30 mL
    e. 40 mL
12. Major components of the epididymis include:
    a. caput, corpus, cauda
    b. globus minor, ductus deferens, cauda

c. rete testis, efferent ductules, caput
d. efferent ductules, caput, cauda
e. septa, efferent ductules, corpus

13. Which of the following is characteristic of meiosis?
    a. Cellular DNA content doubles during the first (meiotic) cell division.
    b. Chromosome number is maintained
    c. Homologous chromosomes (non-sister) pair during the second cell division
    d. Crossing over is present
    e. Centromeres do not divide
14. When does a testosterone peak occur during a human male's life?
    a. 2 months
    b. 6 to 9 months
    c. 30 to 40 years
    d. 50 to 60 years
    e. >70 years
15. With chronic vasal obstruction, the highest sperm motility is found in what region of the epididymis?
    a. Rete testis
    b. Efferent ducts
    c. Proximal epididymis
    d. Mid epididymis
    e. Distal epididymis
16. Which of the following occurs during spermiogenesis?
    a. Type B spermatogonia divide to form primary spermatocytes
    b. Cellular remodeling within the basal compartment of the seminiferous tubule
    c. DNA replication
    d. Nuclear compaction
    e. Cytokinesis
17. Changes to sperm during epididymal transit include:
    a. increased positive charge.
    b. sulfhydryl reduction.
    c. decreased phospholipid content.
    d. reduced membrane rigidity.

e. increased capacity for glycolysis.

18. Which of the following germ cell types is considered a true stem cell?
    a. Elongating spermatids
    b. Primary spermatocytes
    c. Round spermatids
    d. Secondary spermatocytes
    e. Type A spermatogonia
19. During their development within the male reproductive tract prior to ejaculation, sperm spend the majority of time in which organ:
    a. epididymis
    b. ejaculatory ducts
    c. seminal vesicle
    d. testicle
    e. vas deferens
20. The sperm region containing mitochondria is the:
    a. tail.
    b. head.
    c. acrosome.
    d. midpiece.
    e. axoneme.
21. Which of the following statements about testosterone is most accurate?
    a. It is produced by the exocrine testis
    b. It exists mainly in the unbound or “free” form in the circulation
    c. It is regulated by FSH
    d. Its production is increased by excess prolactin
    e. Its metabolite is dihydrotestosterone (DHT)
22. Spermiogenesis includes all of the following processes EXCEPT:
    a. loss of cytoplasm
    b. formation of the acrosome
    c. flagellar formation
    d. migration of cytoplasmic organelles
    e. cell division
23. After ejaculation the contents of the vas deferens are:

a. returned to the seminal vesicles
b. maintained in the ampulla
c. released into the ejaculatory ducts
d. propelled back into the epididymis
e. released into the bladder

Answers

1. **b. Wolffian ducts.** The müllerian ducts regress in the male. The indifferent gonad migrates to the urogenital ridge to become the testicle. The gubernaculum testis is responsible for pulling the testis into the scrotum during development.
2. **d. 70% to 80%.** In humans, interstitial tissue takes up 20% to 30% of the total testicular volume.
3. **d. FSH, testosterone.** FSH and testosterone play the most important role in the regulation of Sertoli cell function. The effect of prolactin and estradiol on Sertoli cells is not well defined. ABP is made by Sertoli cells.
4. **c. Seminal vesicles.** At least 65% to 70% of ejaculate volume is derived from the seminal vesicles, with the remainder from the vas deferens (with sperm) and prostatic secretions. Periurethral glands may also contribute a small amount of fluid to the normal ejaculate.
5. **e. Pudendal artery.** The arterial supply to the human testis and epididymis is derived from three sources: the internal spermatic artery, the deferential artery, and the external spermatic or cremasteric artery.
6. **c. Low, because exogenous testosterone inhibits pituitary production of FSH and LH.** Because of negative feedback inhibition that maintains homeostatic balance in the pituitary-gonadal axis, exogenous testosterone of any type will cause anterior pituitary production of LH and FSH to fall. This results in azoospermia in most of men on anabolic steroids, but the effect will vary based on the dose, frequency, and duration of the steroid regimen

7. **a. Sertoli-Sertoli cell junctions.** Sertoli-Sertoli tight junctions prevent the deep penetration of electron-opaque tracers into the seminiferous epithelium from the testicular interstitium. These junctions between Sertoli cells also segregate premeiotic germ cells (spermatogonia) from meiotic and postmeiotic germ cells and are the structural basis for the "blood-testis" barrier.
8. **a. Inhibits gonadotropin-releasing hormone (GnRH) and LH.** States of hyperprolactinemia act to inhibits GnRH pulses and also LH pulses, which is turn results in lower testosterone production.
9. **e. Spermatogonia, spermatocyte, spermatid.** Proceeding from the least to the most differentiated, they are named dark type A spermatogonia (Ad); pale type A spermatogonia (Ap); type B spermatogonia (B); preleptotene primary spermatocytes (R); leptotene primary spermatocytes (L); zygotene primary spermatocytes (z); pachytene primary spermatocytes (p); secondary spermatocytes (II); and Sa, Sb1, Sb2, Sc, Sd1, and Sd2 spermatids.
10. **a. Testosterone.** Inhibin and activin feedback on FSH secretion. Prolactin in excess can downregulate LH secretion.
11. **c. 20 mL.** Although ethnic and racial variations exist, 18 to 22 mL is considered normal testis volume for adult human males.
12. **a. Caput, corpus, cauda.** Anatomically, the epididymis is divided into three regions: the caput or head, the corpus or body, and the cauda or tail of the epididymis. On the basis of histologic criteria, each region can be subdivided into distinct zones separated by transition segments.
13. **d. Crossing over is present.** Crossing over of sister chromatids and exchange of genetic material is the hallmark of chromosomal recombination during meiosis. This does not occur in mitosis. Two other unique characteristics of meiosis are that homologous chromosomes pair during meiosis I, the first cell division, and chromosome numbers in daughter cells are half (haploid) that of mitotic cells.
14. **a. 2 months.** A testosterone peak occurs at approximately 2 months of age. Another peak occurs earlier, during the first trimester of gestation and a third peak occurs later at puberty.

15. **c. Proximal epididymis.** Studies in men with congenital absence of the vas deferens or epididymal obstruction from vasectomy frequently report poor motility in sperm aspirated from the distal epididymis, with optimal sperm quality and motility in the proximal epididymis.
16. **d. Nuclear compaction.** In spermiogenesis, cell division or cytokinesis does not occur. In addition, spermiogenesis occurs in the adluminal compartment and not the basal compartment. Spermatids and not spermatogonia are the cell types undergoing the metamorphosis characteristic of spermiogenesis.
17. **e. Increased capacity for glycolysis.** Sperm undergo numerous metabolic changes during epididymal transit. Animal studies describe the acquisition of an increased capacity for glycolysis, changes in intracellular pH and calcium content, modification of adenylate cyclase activity, and alterations in cellular phospholipid and phospholipid-like fatty acid content.
18. **e. Type A spermatogonia.** Type A spermatogonia are the only true stem cell in the testis as they can either self renew or differentiate down the spermatogenic pathway.
19. **d. Testicle.** Sperm spend 45 to 60 days developing in the testis and 2 to 12 days in the epididymis. Depending on ejaculation frequency, they can also be stored in the cauda epididymis from for days to several weeks.
20. **d. Midpiece.** The middle piece of the sperm is a highly organized segment consisting of helically arranged mitochondria surrounding a set of outer dense fibers and the characteristic 9 + 2 microtubular structure of the axoneme.
21. **e. Its metabolite is dihydrotestosterone (DHT).** Testosterone is converted to DHT in target tissue by 5α-reductase. Testosterone is produced by the endocrine testis, exists mainly in bound forms in plasma, and is mainly regulated by LH, not FSH. Excess prolactin decreases testosterone production.
22. **e. Cell division.** During spermiogenesis, the products of meiosis, the round Sa spermatids, metamorphose into mature spermatids. During this process, extensive changes occur in both the spermatid cytoplasm and the nucleus, but cell division is not required. These

changes include loss of cytoplasm, formation of the acrosome, formation of the flagellum, and migration of cytoplasmic organelles to positions characteristic of the mature sperm.

23. **d. Propelled back into the epididymis.** Studies have shown that after sexual stimulation or ejaculation, the contents of the vas deferens are propelled back toward the cauda epididymis, because the distal portion of the vas deferens contracts with greater amplitude, frequency, and duration than does the proximal portion.

Chapter Review

1. Normal testosterone and sperm production depends on the pulsatile secretion of hypothalamic GnRH and LH and FSH from the anterior pituitary gland.
2. Regulation of hypothalamic-pituitary-gonadal (HPG) axis hormones occurs primarily through negative feedback.
3. The determination of maleness is derived from the *SRY* gene on the Y chromosome. However, developmental genes such as *WNT4* and *DAX1* are considered antitestis genes and can proactively induce female gonadal development.
4. Changes to the HPG axis with paternal age include lower testosterone levels, blunted axis feedback, and irregular hormone pulsatility.
5. The testis contains 250 m of seminiferous tubules and 700 million Leydig cells in the young adult.
6. Spermatogenesis occurs in stages, cycles, and waves to ensure constant sperm production.
7. Genes on the X as well as the Y chromosome govern spermatogenesis and contribute to male infertility.
8. With paternal age, there are increases in sperm structural chromosomal abnormalities, autosomal dominant mutations, and epigenetic alterations leading to disease in offspring.
9. The epididymis consists of principal cells with absorptive and secretory function, basal cells derived from macrophages, and contractile cells that facilitate sperm transport.

10. During epididymal passage, sperm mature by gaining progressive motility and the ability to bind to and penetrate the egg zona pellucida.
11. Epididymal function is temperature and androgen (mainly DHT) dependent, important considerations for cryptorchidism, varicocele, and 5α-reductase use.
12. The vas deferens is of wolffian (mesonephric) duct origin and serves to transport sperm from the cauda epididymis to the ejaculatory duct during seminal emission.
13. The seminal vesicle and ejaculatory duct unit is analogous to the bladder and urethra and is subject to both physical blockage and functional disorders that result in infertility.
14. Sperm are ciliated cells that possess a "9 + 2" axonemal structure that allows motility.
15. It is estimated that 200 to 300 genes regulate sperm motility.
16. Sperm motility defects, termed ciliary dyskinesias, are common and can be either correctable (nonspecific flagellar anomalies) or genetic (dysplasia of the fibrous sheath).
17. Human sperm mtDNA is a circular, histone- and intron-free DNA ring that encodes for respiratory-chain–complex proteins and is very susceptible to mutations.

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Integrated Men's Health

Androgen Deficiency, Cardiovascular Risk, and Metabolic Syndrome

Neil Fleshner, Miran Kenk, and Steven Kaplan

Keywords

bladder cancers; cardiovascular risk; erectile dysfunction; gender differences; men's health; mental health; metabolic syndrome; prostate cancer; testosterone supplementation

Questions

1. While present throughout the world, the region showing the largest gap in life expectancy between genders is:
    a. North America
    b. Southeast Asia
    c. Western Europe
    d. Eastern Europe
    e. sub-Saharan Africa
2. All of the following are proposed as factors contributing to the gender gap in life expectancy, EXCEPT:
    a. Increased exposure of males to physical and environmental harm in the workplace
    b. Higher propensity for risk-taking behaviors in males

c. Masculinity-defined norms of health behavior
d. Improved postpartum care
e. Lower compliance with treatment in male patients

3. All of the following factors are classical components of metabolic syndrome, EXCEPT:
    a. Hypertension
    b. Abdominal obesity
    c. Insulin resistance/hyperglycemia
    d. Dyslipidemia
    e. Lower urinary tract symptoms
4. Which of the following statements regarding the physiology of metabolic syndrome is TRUE?
    a. Obesity, adipocyte hypertrophy, and adipocyte hyperplasia lead to systemic suppression of inflammatory processes
    b. Insulin resistance results in the suppression of pancreatic beta cells, leading to diminished insulin secretion and reduced circulating insulin levels
    c. Insulin resistance results in decreased very-low-density lipoprotein (VLDL) production and increased VLDL clearance
    d. Insulin resistance and hyperinsulinemia lead to sympathetic nervous system hyperactivity
    e. Genetics play a major role in the etiology of metabolic syndrome.
5. The following urological conditions are linked with metabolic syndrome, EXCEPT:
    a. Kidney stones
    b. Bladder cancer
    c. Overactive bladder
    d. Priapism
    e. Prostate cancer
6. Which of the following statements regarding metabolic syndrome is TRUE?
    a. Obesity is defined as BMI >20 kg/m^2 in adults.
    b. Dietary or lifestyle modifications such as physical fitness may impact both incidence and outcomes of urological diseases.

c. Exercise decreases insulin-mediated glucose uptake and stimulates inflammation.
d. Dietary change, weight loss, and exercise exacerbate erectile dysfunction in majority of patients.
e. There is no clinical evidence showing that exercise programs improve quality-of-life factors.

7. Which of the following statements is FALSE?
   a. Reduction of circulating testosterone levels occurs as part of normal aging.
   b. Testosterone supplementation is available in formulations for oral, intramuscular, transdermal, sublingual, or transnasal routes.
   c. Clinical data suggest a modest impact of testosterone supplementation on physical functioning.
   d. Globally, testosterone prescribing has been decreasing over the past decades.
   e. Large-scale placebo-controlled prospective trials following patients over decades are needed to establish the safety and benefits of testosterone supplementation.
8. In making decisions regarding testosterone replacement therapy, the clinician should do all of the following, EXCEPT:
   a. Limit diagnosis of testosterone deficiency to symptomatic males with unequivocally low testosterone levels.
   b. Evaluate the patient's history and risk of cardiovascular events.
   c. Discuss potential risks of testosterone replacement therapy.
   d. Evaluate patient's history or risk of prostate cancer.
   e. Recommend testosterone supplementation in men who are planning fertility in the near term.
9. Testosterone replacement should not be recommended in:
   a. men planning fertility in the near term.
   b. men with breast or prostate cancer, or at significant risk of prostate cancer.
   c. men with elevated hematocrit.
   d. men with uncontrolled heart failure, myocardial infarction, or stroke within the last 6 months.

e. all of the above.

10. The following predisposing risk factors and pathological processes apply to both cardiovascular disease and erectile dysfunction, EXCEPT:
    a. Dyslipidemia
    b. Inflammation
    c. Atherosclerosis
    d. Hypotension
    e. Type 2 diabetes mellitus
11. Which of the following is FALSE of the relationship between erectile dysfunction and cardiovascular disease?
    a. Erectile dysfunction could be an early warning of elevated risk of sudden cardiovascular events.
    b. Severity and duration of erectile dysfunction has been correlated with the magnitude of coronary disease.
    c. There is a clear and defined temporal relationship between erectile dysfunction and cardiovascular disease.
    d. Coronary artery calcium score can serve as a marker of atherosclerosis in arteries, including penile arteries.
    e. Erectile dysfunction and cardiovascular disorders are increasingly considered to be independent predictors of each other.
12. Which of the following statements about mental health is FALSE?
    a. Suicide rate in women is four times higher than in men.
    b. An equal number of men and women develop bipolar disorder.
    c. Ninety percent of people diagnosed with schizophrenia before the age of 30 are men.
    d. Men predisposed to suicide are those that are middle-aged and of lower socioeconomic class.
    e. Over 6 million men in the United States suffer from depression.
13. Which of the following is TRUE of opioid abuse?
    a. Prevalence of opioid abuse has been decreasing in recent years.
    b. Physicians prescribing opioids are encouraged to initially prescribe the lowest effective dose.

c. Chronic opioid use stimulates hypothalamic-pituitary-gonadal axis.
d. Opioid-related endocrine dysfunction could drive some chronic opioid users to seek formal treatment.
e. b and d.

14. General categories of a comprehensive approach to integrative men's health should include all of the following, EXCEPT:
    a. Conditions that are unique to men.
    b. Diseases or illnesses that are more prevalent in men compared to women.
    c. Health issues that are not covered by other specialties (e.g., cardiology, endocrinology).
    d. Health issues for which risk factors and adverse outcomes are different in men.
    e. Health issues for which different interventions to achieve improvements are required for men.

Answers

1. **d. Eastern Europe.** Globally, average life expectancy is longer in women compared to men. While this difference in longevity is observed in all regions of the world (regardless of the level of industrial development), it is most pronounced in Eastern Europe. Women in this region tend to live approximately 7 years longer than men.
2. **d. Improved postpartum care.** A series of factors have been identified that place men at higher risk of death and disease. These include increased exposure to physical and environmental harm in the workplace, propensity for risk-taking behaviors, and masculinity defined norms of health behavior that may negatively impact both acute and chronic illness-related outcomes.
3. **e. Lower urinary tract symptoms.** The WHO, American Association of Clinical Endocrinologists (AACE), European Group for the Study of Insulin Resistance (EGIR), Adult Treatment Panel III (ATP III), and International Diabetes Federation definitions of

metabolic syndrome comprise different criteria for defining metabolic syndrome. Classic definition includes hypertension, abdominal obesity, dyslipidemia, and insulin resistance (or hyperglycemia). Lower urinary tract symptoms (LUTS) have been linked with metabolic syndrome by epidemiologic data but are not included in its definition.

4. **d. Insulin resistance and hyperinsulinemia lead to sympathetic nervous system hyperactivity.** Metabolic syndrome includes a number of physiological alterations, including abdominal obesity, insulin resistance, dyslipidemia, and hypertension. Insulin resistance and hyperglycemia are associated with increased stimulation of pancreatic beta cells to secrete more insulin, resulting in hyperinsulinemia. Insulin resistance and hyperinsulinemia lead to hyperactivation of the sympathetic nervous system and hypertension.
5. **d. Priapism.** Metabolic syndrome was classically associated with elevated risk of cardiovascular disease, diabetes, and stroke. Urological conditions are increasingly recognized to be highly prevalent among men with metastatic syndrome. Among these are renal conditions (such as renal insufficiency, urolithiasis, and renal tumors), bladder cancers, overactive bladder, lower urinary tract symptoms, prostate cancer, low testosterone levels, and erectile dysfunction.
6. **b. Dietary or lifestyle modifications such as physical fitness may impact both incidence and outcomes of urological diseases.** Frequent physical exercise reduces the risk of many chronic diseases. Exercise increases insulin-mediated glucose uptake and decreases inflammation by reducing circulating levels of insulin-like growth factor (IGF-1). In addition to the meta-analyses showing benefit of exercise for reducing prostate cancer, a growing body of data links inactivity with worse disease outcomes.
7. **d. Globally, testosterone prescribing has been decreasing over the past decades.** Testosterone has been approved and made available for treatment of hypogonadal males, including men with testicular trauma and those with rare bilateral tumors. While testosterone levels decrease as part of aging, the benefits and safety

profile of exogenous testosterone supplementation have not been established. Despite the lack of conclusive data (requiring large-scale, placebo-controlled trials), use of testosterone supplements has increased significantly over the past years, with prescriptions for testosterone replacement therapy dramatically increasing in the UK and USA.

8. **e. Recommend testosterone supplementation in men who are planning fertility in the near term.** The lack of conclusive clinical trial data makes it essential that the practicing clinician understands the controversies and is able to communicate them to her/his patients. Endocrine Society guidelines published in 2018 (a) advised the clinicians to limit the diagnosis of testosterone deficiency to symptomatic males with unequivocally low testosterone levels; (b) recommend against testosterone replacement therapy in men planning fertility in the near term and those with conditions that could be exacerbated by testosterone; and (c) recommend discussing the potential risks (including cardiovascular) associated with testosterone-replacement therapy.
9. **e. All of the above.** According to the guidelines of the Endocrine Society, testosterone should not be prescribed to men with breast or prostate cancer, those planning fertility in the near term, or in those with significant prostate cancer risk, elevated hematocrit, untreated severe obstructive sleep apnea, severe lower urinary tract symptoms, uncontrolled heart failure, myocardial infarction, or stroke within the last 6 months, or thrombophilia.
10. **d. Hypotension.** There are a host of predisposing risk factors and underlying pathophysiologic processes for both cardiovascular disease (CVD) and erectile dysfunction (ED). These include, but are not limited to dyslipidemia, smoking, hypertension, and T2DM. The underlying mechanisms of both CVD and ED include inflammation, atherosclerosis, and endothelial dysfunction.
11. **c. There is a clear and defined temporal relationship between erectile dysfunction and cardiovascular disease.** There is emerging consensus that erectile dysfunction and cardiovascular disease are independent risk factors for each other. However, the temporal relationship between the two disorders is less clear. A

landmark study, The Multi-Ethnic Study of Atherosclerosis (MESA), demonstrated that subclinical cardiovascular disease is a predictor of erectile dysfunction and potentially of subsequent major adverse cardiac events. Earlier studies suggest that there is a 2- to 3-year interval between the onset of erectile dysfunction and cardiovascular symptoms. Since about 50% of men with sudden cardiovascular events may have had no previous symptoms, ED may be the single warning of the elevated risk of sudden cardiovascular events.

12. **a. Suicide rate in women is four times higher than in men.** Mental illnesses, including anxiety, depression, and suicide, are becoming increasingly prevalent, while concomitantly being underdiagnosed, in men. Specifically, over 6 million men suffer from depression including symptoms such as fatigue, irritability, and loss of interest in work or hobbies. High rates of mental illness have led to a significant increase in incidences of both suicide and substance abuse. Highest prevalence of suicide in the United States occurred in people between ages 45 and 54, with men exhibiting a suicide rate four times higher than women.
13. **b. Physicians prescribing opioid are encouraged to initially prescribe the lowest effective dose.** In recent years the emotional and economic consequences of opioid misuse on public health has increased dramatically and is now understood as a major source of morbidity and mortality in the United States. Despite a decrease in the number of total opioids prescribed in the U.S. from a peak in 2012, some 33,000 people died of opioid overdoses in 2015 (including 15,000 from prescription sources), and currently more than 115 Americans die daily after overdosing on opioids. There appears to be a high degree of variability in the dispensing of pain medications after surgery and there are some who routinely dispense above recommended safety thresholds. Urologists need to foster greater awareness of the potential harms associated with even a short course of opioids. Those prescribing should adopt responsible prescribing practices, such as initially prescribing the lowest effective dose and offering repeat assessments as needed.

14. **c. Health issues that are not covered by other specialties (e.g., cardiology, endocrinology).** A men's health center should be focused on conditions, diseases, and health issues that exhibit higher impact on men, or that require unique strategies in men. A men's health curriculum must include contributions from a host of medical specialties. Optimally, such a center should include partnerships with various subspecialists in cardiology, endocrinology, psychiatry, orthopedics, and dermatology.

Chapter Review

1. Men throughout the world live shorter lives than women throughout the world.
2. The gender gap persists despite improvements in overall life expectancy at birth.
3. Men not only live shorter lives than women, they also live sicker lives.
4. Gender differences in illness-related outcomes reflect social, environmental, and physiological factors.
5. Innovative and coordinated programs targeting men are required to promote social and behavioral changes that would improve men's health.
6. Metabolic syndrome, defined as a combination of dyslipidemia, glucose insensitivity, and obesity is increasingly prevalent around the world.
7. In addition to its association with increased CVD risks, metabolic syndrome is linked with pelvic health issues, including lower urinary tract symptoms, ED, and hypogonadism, as well as genitourinary tumors (kidney, bladder, prostate).
8. Dietary or lifestyle modifications may improve outcomes in many of these urologic diseases, with clinical trials currently testing metabolic modulators statins and metformin.
9. Exercise-based interventions in men with advanced prostate cancer improve quality of life.

10. Dietary change, weight loss, and exercise may improve erectile function.
11. Men's health should include individualized and precision-based intervention programs to alter the arc of pelvic conditions.
12. Testosterone replacement therapy in countering low testosterone levels that occur as part of the aging process is controversial due to inadequately elucidated effectiveness and safety profile.
13. Recent large-scale randomized trials demonstrated some improvements in sexual and physical function and vitality.
14. Limited clinical trial evidence and retrospective cohort analyses suggest that testosterone replacement therapy increases the risk of cardiovascular events.
15. Practicing urologists should keep current with the recently updated guidelines and current research on testosterone-replacement therapy to be able to discuss the risks and benefits for individual patients.
16. There are a number of overlapping risk factors for both CVD and ED.
17. The temporal relationship between ED and subclinical CVD remains to be determined.
18. CAC score can serve as a "disease score" and a surrogate for accelerated atherosclerosis process in arteries, including penile arteries, with implications for vascular ED.
19. Mental health issues, including anxiety, depression, and suicide, are increasingly prevalent in men while concomitantly being under diagnosed.
20. Over 6 million men in the United States suffer from depression, including symptoms such as fatigue, irritability, and loss of interest in work or hobbies.
21. Men are at higher risk for suicide, with a rate approximately four times higher than that of women.
22. Opioid abuse is now a leading health care crisis, with significant morbidity and mortality.
23. The opioid epidemic is notable for the expansiveness of the problem in the United States, spanning divisions of gender, race, age, and income level.

24. Clinicians who treat hypogonadism should be cognizant of the relationship between chronic opioid and symptomatic hypogonadism.

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Male Infertility

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Keywords

assisted reproduction; infertility; semen analysis; sperm; sperm cryopreservation

Questions

1. For couples practicing optimal timing methods to conceive, the proportion who conceive after six cycles should be approximately:
    a. 1/2.
    b. 1/3.
    c. 1/5.
    d. 2/5.
    e. 4/5.
2. For couples desiring to conceive, around the time of ovulation, they should have intercourse:
    a. daily.
    b. twice daily.
    c. every other day.
    d. in the morning.
    e. near noon.
3. In the industrialized world, female fecundity typically declines most rapidly after:
    a. puberty.

b. age 30 years.
c. age 35 years.
d. age 40 years.
e. menopause.

4. All of the following medications are associated with male reproductive dysfunction EXCEPT:
    a. cimetidine.
    b. spironolactone.
    c. indinavir.
    d. prednisone.
    e. lisinopril.
5. A man with oligoasthenospermia has inflammatory bowel disease and is prescribed sulfasalazine. He should be counseled to:
    a. repeat a semen analysis.
    b. continue sulfasalazine.
    c. substitute sulfasalazine with colchicine.
    d. substitute sulfasalazine with mesalazine.
    e. discontinue all medications.
6. A 30-year-old man is attempting to conceive with his 28-year-old wife. She is diagnosed with premature ovarian failure, and hematoxylin and eosin microscopic staining of his sperm is consistent with the presence of *Escherichia coli.* His bulk seminal parameters are otherwise unremarkable. You recommend:
    a. swim up and intrauterine insemination (IUI).
    b. a 3-week course of ciprofloxacin, 500 mg twice daily.
    c. prostatic culture.
    d. seminal vesicle aspiration for culture.
    e. observation.
7. The effect of human immunodeficiency virus (HIV) infection on bulk seminal parameters is:
    a. none.
    b. decreased motility.
    c. increased number of morphologically abnormal forms.
    d. decreased concentration.
    e. decreased volume.

8. Direct exposure of the testis to ionizing radiation causes irreparable damage to spermatogenesis at doses at and above:
    a. 2.5 Gy.
    b. 5 Gy.
    c. 7.5 Gy.
    d. 10 Gy.
    e. 20 Gy.
9. The typical differential between core body and scrotal temperature is:
    a. 0.5°C.
    b. 0.5°F.
    c. 1°C to 2°C.
    d. 2°C to 4°C.
    e. 6°C.
10. An 18-year-old man has bilateral testes palpable at the external inguinal rings. He desires to father children in the future. Two centrifuged semen analyses reveal azoospermia. Serum testosterone is 410 ng/dL, and follicle-stimulating hormone is 22 IU/L. The best next step is:
    a. counsel that when he is ready, he and his partner should use donor sperm or adopt.
    b. clomiphene citrate, 50 mg every other day.
    c. testis biopsy.
    d. microsurgical testis sperm extraction with cryopreservation if sperm are found.
    e. bilateral orchidopexy.
11. The lubricant that does NOT affect sperm function or DNA integrity is:
    a. Astroglide.
    b. K-Y Jelly.
    c. Pre-Seed.
    d. Replens.
    e. saliva.
12. The serum hormonal pattern commonly observed in male obesity is:

	Testosterone	Sex Hormone–Binding Globulin (SHBG)	Estradiol
a.	↓	↓	↓
b.	↓	↓	↑
c.	↓	↑	↓
d.	↓	↑	↑
e.	↑	↑	↑

13. The finding most consistent with spermatogenic impairment is:
    a. a testis volume as measured by Prader orchidometer less than 30 mL.
    b. a testis volume as measured by scrotal ultrasonography less than 30 mL.
    c. a testis longitudinal axis as measured by caliper orchidometer less than 4.6 cm.
    d. an engorged epididymis.
    e. an absent vas deferens.
14. A man evaluated for azoospermia has absent vasa bilaterally. The next step is cystic fibrosis transmembrane conductance regulator (CFTR) gene sequence assay for him and his partner and:
    a. transrectal ultrasound.
    b. renal ultrasound.
    c. postejaculatory urinalysis.
    d. seminal vesicle aspiration.
    e. testis biopsy.
15. The following condition increases sex hormone–binding globulin (SHBG):
    a. aging
    b. diabetes mellitus
    c. hypothyroidism
    d. obesity
    e. testosterone therapy
16. A man is found to be azoospermic. The following most strongly suggests spermatogenic dysfunction as a cause rather than obstruction:

a. testis long axis on physical examination 4.2 cm, FSH assay 4 IU/L
b. testis long axis on physical examination 4.3 cm, FSH assay 10 IU/L
c. bioavailable testosterone, 166 ng/dL, FSH assay 6 IU/L
d. luteinizing hormone (LH), 2 IU/L, FSH 2 IU/L
e. testosterone, 160 ng/dL, LH 9 IU/L

17. A reasonable threshold for sperm concentration above which a man can be considered fertile is:
    a. 15 million/mL.
    b. 20 million/mL.
    c. 48 million/mL.
    d. 80 million/mL.
    e. 200 million/mL.
18. Brown-hued semen is often associated with:
    a. ingested asparagus.
    b. ejaculatory ductal obstruction.
    c. sexual activity.
    d. spinal cord injury.
    e. urinary tract infection.
19. The optimal time in days of abstinence to wait after an ejaculate for a semen analysis is:
    a. 1.
    b. 2.
    c. 3.
    d. 5.
    e. 7.
20. A man who has been unsuccessful in impregnating his wife during the past year is identified to have azoospermia and semen volumes of 0.8 and 0.5 mL. The next step is:
    a. testis biopsy.
    b. scrotal ultrasound.
    c. postejaculatory urinalysis.
    d. clomiphene citrate, 50 mg every other day.
    e. transurethral resection of ejaculatory ducts.

21. A 32-year-old man presents for fertility evaluation. Physical exam and endocrine assessment are normal. Semen analysis demonstrates volume 2.5 mL, density 84 million/mL, motility 71%, and strict morphology 0% with a variety of abnormal forms. The next step is:
    a. vitamin E.
    b. reassurance.
    c. testis biopsy.
    d. scrotal ultrasound.
    e. antisperm antibody assay.
22. Two semen analyses from a man undergoing a reproductive evaluation include volumes of 2.5 and 1.8 mL, densities of 24 and 28 million/mL, and motility 0%. The next step is:
    a. vital stain.
    b. immunobead assay.
    c. repeat semen analysis.
    d. sperm chromatin structure assay (SCSA).
    e. computer-assisted semen analysis (CASA).
23. Semen observed under phase contrast light microscopy reveals abundant round cells similar in size and shape to leukocytes. The next step is:
    a. semen culture.
    b. Papanicolaou staining.
    c. transrectal ultrasound.
    d. ciprofloxacin, 1 g daily for 4 weeks.
    e. ibuprofen, 400 mg daily for 2 weeks.
24. A direct assay of sperm-head DNA fragmentation is the:
    a. terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) assay.
    b. acid comet assay.
    c. alkaline comet assay.
    d. sperm chromatin dispersion (SCD) assay.
    e. SCSA.
25. The assay that most closely models incubational in vitro fertilization is:
    a. reactive oxygen species-total antioxidant capacity (ROS-TAC).

b. Penetrak assay.
c. Tru-trax assay.
d. sperm penetration assay.
e. acrosomal assay.

26. A 26-year-old man presenting for fertility evaluation has two semen analyses, one with volume 2.1 mL and density 32 million/mL, and another with volume 2.3 mL and density 28 million/mL. No motile sperm are observed in either, and vital stain demonstrated greater than 80% metabolically active sperm in both. The next step is:
    a. immunobead assay.
    b. pyospermia stain.
    c. electron microscopy.
    d. SCSA.
    e. repeat semen analysis.
27. A 30-year-old man has azoospermia. Physical examination reveals absent vasa bilaterally. CFTR analysis reveals a ΔF508 mutation. The next step is:
    a. repeat CFTR analysis of the man with a 1600-mutation screen.
    b. repeat semen analysis.
    c. CFTR analysis of his wife.
    d. testicular sperm extraction.
    e. testis biopsy.
28. A 28-year-old man desires a biological child. Physical examination is normal. Two semen analyses reveal azoospermia with volumes of 0.8 and 0.7 mL, and no sperm is identified in a postejaculatory urinalysis. FSH is 3.2 IU/L. Transrectal ultrasound reveals seminal vesicles of 0.9 and 1.1 cm in largest diameter, and no midline cyst is visualized. The next step is:
    a. reassurance.
    b. testis biopsy.
    c. repeat semen analysis.
    d. transurethral resection of ejaculatory ducts.
    e. testis sperm extraction and cryopreservation.
29. A 33-year-old man desires a biological child and is found to have azoospermia with semen volumes 2.2 and 2.4 mL on two analyses.

Longitudinal axis of the left testis is 3.3 cm, and the right is 3.5 cm. Serum FSH is 9.3 IU/L, and testosterone is 440 ng/dL. The next step is:

a. testis biopsy.
b. epididymovasostomy.
c. repeat semen analysis.
d. microsurgical testis sperm extraction.
e. clomiphene citrate, 50 mg every other day.

30. A 30-year-old man desires a biological child with his 28-year-old wife. Her evaluation is normal. Semen analysis reveals density 84 million/mL and globozoospermia. The next step is:

a. repeat semen analysis.
b. clomiphene citrate, 50 mg every other day.
c. in vitro fertilization after incubation of sperm with pentoxifylline.
d. intracytoplasmic sperm injection with ejaculated sperm.
e. microsurgical testis sperm extraction.

31. The most common genetic cause of male infertility is:

a. r(Y).
b. 45,X0.
c. 47,XXY.
d. 47,XYY.
e. 46,XY/47,XXY.

32. The biological system responsible for protecting haploid male germ cells from the immune system is:

a. IgG.
b. Prostate specific antigen (PSA).
c. natural killer cells.
d. reactive oxygen species.
e. Sertoli cell tight junctions.

33. A man presents with azoospermia. Testis longitudinal axis is 3.0 cm bilaterally, and the testes are soft. Laboratory evaluation reveals testosterone 140 ng/dL, LH less than 1.0 IU/L, and FSH less than 1.0 IU/L. The next step is:

a. repeat semen analysis.
b. transrectal ultrasound.

c. microsurgical testis sperm extraction.
d. smell test.
e. Y-chromosomal microdeletion assay.

34. A man presents with erectile dysfunction and azoospermia. Testis longitudinal axis is 5 cm bilaterally, and the testes are firm. Laboratory evaluation reveals bioavailable testosterone 150 ng/dL and prolactin 42 μg/L. The next step is:
    a. repeat prolactin.
    b. bromocriptine, 2.5 mg orally per day.
    c. cabergoline, 0.25 mg orally twice a week.
    d. clomiphene citrate, 50 mg orally every other day.
    e. cranial magnetic resonance imaging.
35. Laboratory values associated with androgen receptor insensitivity include:
    a. significantly elevated testosterone, LH, and FSH.
    b. mildly elevated testosterone, LH, and FSH.
    c. significantly elevated testosterone, mildly elevated LH, and normal FSH.
    d. decreased testosterone, and elevated LH and FSH.
    e. decreased testosterone, LH, and FSH.
36. The most common mutation in the cystic fibrosis transmembrane conductance region, or CFTR, is:
    a. R117H.
    b. R334W.
    c. R347P.
    d. ΔF508.
    e. G542X.
37. A man presents with a palpable right-sided varicocele of 3 months onset and no varicocele palpable on the left side. Testis longitudinal axis is 5 cm bilaterally, and a semen analysis reveals volume 2.0 mL, density 33 million/mL, and motility 38%. The next step is:
    a. semen analysis with strict morphology.
    b. renal ultrasound.
    c. CFTR testing.
    d. right varicocelectomy.
    e. bilateral varicocelectomy.

38. Electroejaculation is planned for a man with anejaculation due to spinal cord injury at T5. Treatment should include monitoring and therapy before the procedure with oral:
    a. lisinopril.
    b. nifedipine.
    c. pseudoephedrine.
    d. captopril.
    e. ciprofloxacin.
39. A couple desires children. Semen analysis reveals volume 2.2 mL, density 58 million/mL, and motility 18%, and all sperm have enlarged heads with multiple tails. The next step is:
    a. testis biopsy.
    b. scrotal ultrasound.
    c. repeat semen analysis.
    d. IUI with donor sperm.
    e. in vitro fertilization with intracytoplasmic sperm injection.

Answers

1. **e. 4/5.** Cumulative pregnancy rates in a well-conducted study were 38% at one cycle, 68% at three cycles, 81% at six cycles, and 92% at 12 cycles (Gnoth et al., 2003). [a]
2. **a. Daily.** Although prior recommendations specified intercourse every other day to optimize the probability of conception, a recent study demonstrated that intercourse every day around the time of ovulation is likely the best strategy (Scarpa et al., 2007).
3. **c. Age 35 years.** Whereas women in developing nations may experience a rapid decline in fecundity at a younger age, in the industrialized world, female fecundity declines precipitously after age 35 years (Balasch and Gratacós, 2012).
4. **e. Lisinopril.** All agents listed are spermatotoxins except lisinopril, which may improve bulk seminal parameters (Mbah et al., 2012).
5. **d. Substitute sulfasalazine with mesalazine.** Sulfasalazine is associated with oligoasthenospermia (Stein and Hanauer, 2000). If

sulfasalazine is substituted with mesalazine, adverse effects on sperm are generally reversible (Riley et al., 1987).

6. **e. Observation.** Limited seminal concentrations of the majority of bacteria including *E. coli* have minimal or no effects on sperm motility in vivo (Diemer et al., 2003; Lackner et al., 2006).
7. **a. None.** HIV does not appear to be correlated with a direct negative effect on sperm function (Garrido et al., 2005).
8. **c. 7.5 Gy.** The probability of future fatherhood is significantly decreased with radiation doses to the testes of 7.5 Gy and above (Green et al., 2010). The testis does not need to be directly exposed for adverse spermatogenic effects to occur.
9. **d. 2°C to 4°C.** Unlike female gonads, the testes are extracorporeal and subject to thermal regulation by a vascular heat exchange mechanism and muscular activity controlling proximity to the body, resulting in a scrotal temperature maintained between 2°C and 4°C below body core temperature (Setchell, 1998; Thonneau et al., 1998).
10. **e. Bilateral orchidopexy.** The most significant feature affecting this man's reproductive potential is bilateral cryptorchidism. Although the likelihood of improving fertility is hampered by his advanced age, no value will be derived from waiting. Testis biopsy is unnecessary, and his total testosterone is adequate. In this patient with distal cryptorchidism who does not desire offspring presently, removing the spermatotoxic insult of cryptorchidism is the most prudent course.
11. **c. Pre-Seed.** Nearly all lubricants are spermatotoxic, including saliva. In a study of a variety of lubricants, Pre-Seed did not result in a significant decrease in sperm motility or chromatin integrity (Agarwal et al., 2008).
12. **Row b.** In obese males, serum testosterone is decreased (Hammoud et al., 2006). SHBG is typically reduced, likely because of increased circulating insulin (Hammoud et al., 2006; 2008; Pauli et al., 2008; Teerds et al., 2011). Estradiol is increased because of peripheral conversion from testosterone by an overabundance of adipose cells containing the enzyme aromatase (Aggerholm et al.,

2008; Chavarro et al., 2010; Hammoud et al., 2006, 2010; Hofny et al., 2010).

13. **c. A testis longitudinal axis as measured by caliper orchidometer less than 4.6 cm.** A measurement of the long axis of the testis of 4.6 cm or less is associated with spermatogenic impairment (Schoor et al., 2001).
14. **b. Renal ultrasound.** Renal agenesis is noted in 11% of men with congenital bilateral absence of the vas deferens (Schlegel et al., 1996).
15. **a. Aging.** Concentration of SHBG increases with age, resulting in decreased bioavailable testosterone (Bhasin et al., 2010).
16. **b. Testis long axis on physical examination 4.3 cm, FSH assay, 10 IU/L.** If the FSH assay result is 7.6 IU/L or less and the testis long axis is greater than 4.6 cm, the probability of obstruction is 96%; conversely, if the FSH values is greater than 7.6 IU/L and the testis long axis 4.6 cm or less, the probability that azoospermia is due to spermatogenic dysfunction is 89% (Schoor et al., 2001).
17. **c. 48 million/mL.** By classification and regression tree analysis, or CART, one large study demonstrated that for sperm concentration, 13.5 million/mL was found to be the lower parameter below which IUI success would be unlikely, and 48.0 million/mL was identified as the upper parameter above which IUI outcomes were favorable (Guzick et al., 2001).
18. **d. Spinal cord injury.** The ejaculate is normally white or light gray. A brown hue is often observed in spinal cord–injured men (Centola, 2011; World Health Organization, 2010).
19. **a. 1.** A single day of abstinence is optimal for assessing bulk seminal parameters (Elzanaty, 2008; Levitas et al., 2005).
20. **c. Postejaculatory urinalysis.** The differential diagnosis of seminal hypovolemia includes retrograde ejaculation, ejaculatory ductal obstruction, and accessory sex gland hypoplasia. The simplest and least invasive test to exclude a diagnosis of retrograde ejaculation is postejaculatory urinalysis, and this should be performed first.
21. **b. Reassurance.** This man's bulk seminal parameters are adequate with the exception of morphology. The variety of abnormal forms excludes rare genetic conditions such as failure of formation of the

acrosomal cap. The assessing technician likely overread abnormal forms, and the patient should be reassured.

22. **a. Vital stain.** In cases of complete asthenospermia, assessment of antisperm antibodies with the immunobead assay is not possible, as it requires some motile sperm. The first diagnosis to exclude is necrospermia, which may be investigated with a vital stain.
23. **b. Papanicolaou staining.** Leukocytes and immature germ cells are not differentiable with light microscopy. A simple stain such as Papanicolaou allows the two to be distinguished (World Health Organization, 2010).
24. **a. Terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) assay.** Direct measures of sperm DNA fragmentation include the TUNEL assay and the comet assay at neutral pH. The other assays listed include denatured DNA (Sakkas and Alvarez, 2010).
25. **d. Sperm penetration assay.** To simulate incubational in vitro fertilization, human sperm are incubated with denuded hamster ova in the sperm penetration assay, or SPA (Margalioth et al., 1986; Yanagimachi et al., 1976).
26. **c. Electron microscopy.** Vital stain excluded necrospermia in this patient with complete asthenospermia. Electron microscopy will identify ultrastructural tail defects if present in the immotile cilia syndrome (Zini and Sigman, 2009).
27. **c. CFTR analysis of his wife.** Should the wife of this azoospermic man with a severe CFTR mutation harbor a severe mutation as well, a child born from surgically extracted sperm and intracytoplasmic sperm injection may be homozygous for a severe mutation and have clinical cystic fibrosis. In order to counsel the couple regarding the probability of that result, CFTR analysis of the wife is indicated.
28. **e. Testis sperm extraction and cryopreservation.** This man's normal physical examination and relatively low FSH indicates a high likelihood of adequate spermatogenesis and obstructive azoospermia. The lack of seminal vesicle dilation suggests that the probability of ejaculatory ductal obstruction is low. The next step is surgical sperm retrieval.

29. **d. Microsurgical testis sperm extraction.** With the FSH assay result greater than 7.6 IU/L and the testes long axis measurements of 4.6 cm or less, the probability that azoospermia is due to spermatogenic dysfunction is 89% (Schoor et al., 2001). The next step is microsurgical testis sperm extraction.
30. **d. Intracytoplasmic sperm injection with ejaculated sperm.** In globozoospermia, the sperm lack acrosomal caps, and sperm heads are rendered spheric rather than ovoid. Without the acrosome, fertilization with sperm in the natural setting or with incubational in vitro fertilization will not be successful. Intracytoplasmic sperm injection is required. Ejaculated sperm is available in this patient, and surgical sperm extraction is consequently unnecessary.
31. **c. 47,XXY.** The presence of a supernumerary X chromosome in 47,XXY, or Klinefelter syndrome, is the most common genetic cause of male infertility (Groth et al., 2013; Oates and Lamb, 2009; Sigman, 2012).
32. **e. Sertoli cell tight junctions.** The haploid male gamete expresses different surface antigens than other diploid cells and is protected from the immune system by tight junctions between Sertoli cells (Walsh and Turek, 2009).
33. **d. Smell test.** Anosmia associated with hypogonadotropic hypogonadism is known as Kallmann syndrome (Kallmann and Schoenfeld, 1944). In a patient with significantly low gonadotropin assay results, the presence of the syndrome is confirmed by a smell test.
34. **a. Repeat prolactin.** Prolactin is a labile assay. Before continuing with further diagnostic assessment or therapy, moderately elevated assay results should first be confirmed with a second test.
35. **c. Significantly elevated testosterone, mildly elevated LH, and normal FSH.** Male infertility associated with androgen receptor insensitivity is characterized by increased testosterone, increased estradiol, increased LH to variable degrees, and typical FSH levels (Sokol, 2009).
36. **d. ΔF508.** The most common CFTR mutation is ΔF508, which is severe (Hampton and Stanton, 2010).

37. **b. Renal ultrasound.** Solitary right varicoceles are rare. Should one be of abrupt onset, renal pathology such as tumor should be considered (Masson and Brannigan, 2014).
38. **b. Nifedipine.** Ejaculatory stimulation for men with spinal cord injuries at a level of T6 or above may result in autonomic dysreflexia, which can be addressed before stimulation by treatment with nifedipine and during the procedure with monitoring of cardiac activity and blood pressure (Brackett et al., 2009; Phillips et al., 2014).
39. **d. IUI with donor sperm.** Because of the high rate of aneuploidy in sperm associated with macrocephaly and multiple tails, intracytoplasmic sperm injection with biological gametes is not recommended (Machev et al., 2005; Perrin et al., 2012; Sun et al., 2006).

Chapter Review

1. Human spermatogenesis requires 64 days to complete and 5 to 10 days of epididymal transit time.
2. 5-α reductase inhibitors have a limited effect on spermatogenesis.
3. Cannabis decreases plasma testosterone; heavy alcohol use increases the conversion of testosterone to estradiol.
4. DNA damage can be detected up to 2 years following chemotherapy.
5. Following torsion of the testis, 11% of men develop antisperm antibodies.
6. Testis size correlates well with sperm production.
7. Patients with bilateral absence of the vas should be evaluated for a cystic fibrosis gene mutation.
8. Varicoceles are present in 15% of the adult population but occur in 30% to 50% of men presenting with fertility problems.
9. Normal semen volume is between 1 and 5 mL.
10. Progressive motility should be between 32% and 63%.
11. Antisperm antibodies are associated with vasectomy, testes trauma, orchitis, cryptorchidism, testis cancer, and varicocele.

12. Genetic testing should be considered in those with spermatogenic dysfunction causing azoospermia and in those with sperm densities of less than 5 million/mL.
13. The AZF factor is found on the long arm of the Y chromosome; the DAZ genes are located in the AZFc region. A microdeletion of the AZFc region may result in spermatogenic impairment but not necessarily absence of spermatogenesis, but a microdeletion of the AZFa and AZFb regions generally result in absence of spermatogenesis.
14. Spermatogenesis may be highly focal in men with azoospermia so that a random biopsy may miss areas of sperm production.
15. Cumulative pregnancy rates in a well-conducted study were 38% at one cycle, 68% at three cycles, 81% at six cycles, and 92% at 12 cycles.
16. Female fecundity declines precipitously after age 35 years.
17. Limited seminal concentrations of the majority of bacteria, including *E. coli*, have minimal or no effects on sperm motility in vivo.
18. If the FSH assay result is 7.6 IU/L or less and the testis long axis is greater than 4.6 cm, the probability of obstruction is 96%. Conversely, if the FSH value is greater than 7.6 IU/L and the testis long axis 4.6 cm or less, the probability that azoospermia is due to spermatogenic dysfunction is 89%.
19. A sperm concentration of 13.5 million/mL has been found to be the lower parameter below which IUI success is unlikely, and 48.0 million/mL is identified as the upper parameter above which IUI outcomes are favorable.
20. A single day of abstinence is optimal for assessing bulk seminal parameters.
21. The differential diagnosis of seminal hypovolemia includes retrograde ejaculation, ejaculatory ductal obstruction, and accessory sex gland hypoplasia.
22. Leukocytes and immature germ cells are not differentiable with light microscopy. A simple stain such as Papanicolaou allows the two to be distinguished.
23. Direct measures of sperm DNA fragmentation include the terminal deoxynucleotidyl transferase dUTP nick end labeling, or TUNEL.

24. To simulate incubational in vitro fertilization, human sperm are incubated with denuded hamster ova in the sperm penetration assay, or SPA.
25. In globozoospermia, the sperm lack acrosomal caps, and sperm heads are rendered spheric rather than ovoid. Without the acrosome, fertilization with sperm in the natural setting or with incubational in vitro fertilization will not be successful.
26. Klinefelter syndrome is the most common genetic cause of male infertility.
27. Male infertility associated with androgen receptor insensitivity is characterized by increased testosterone, increased estradiol, increased LH to variable degrees, and typical FSH levels.
28. Solitary right varicoceles are rare. Should one be of abrupt onset, renal pathology such as tumor should be considered.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition,* on the Expert Consult website.

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Surgical Management of Male Infertility

Marc Goldstein

Keywords

azoospermia; ICSI; IUI; IVF; male infertility; microsurgery; nonobstructive azoospermia; obstruction; sperm aspiration; testis biopsy; transurethral resection of the ejaculatory ducts; varicocele; vasoepididymostomy; vasogram; vasography; vasovasostomy

Questions

1. Which of the following venous structures are intentionally preserved during varicocelectomy?
    a. External spermatic veins
    b. Internal spermatic veins
    c. Gubernacular veins
    d. Deferential (vasal) veins
    e. Cremasteric veins
2. In the evaluation for vasectomy reversal, which clinical finding is suggestive of epididymal obstruction?
    a. Varicocele
    b. Hydrocele
    c. Sperm granuloma
    d. Normal serum follicle-stimulating hormone (FSH) level
    e. Vasal gap larger than 2 cm

3. Which of the following is NOT an indication for crossed vasovasostomy?
    a. Right: inguinal vas obstruction and normal testis; left: patent vas and atrophic testis
    b. Right: epididymal obstruction, patent vas, and normal testis; left: ejaculatory duct obstruction and normal testis
    c. Right: inguinal vas obstruction and normal testis; left: epididymal obstruction, patent vas, and normal testis
    d. Right: epididymal obstruction and patent right vas above vasectomy site; left: sperm in testicular end of vas and vasectomy site in convoluted vas
    e. Right: congenital absence of epididymis and normal patent vas; left: normal testis and partial absence of vas ending retroperitoneally
4. Compared with the other surgical options for varicocelectomy, the advantages of performing a subinguinal microsurgical varicocelectomy include all of the following EXCEPT:
    a. lower rate of arterial injury.
    b. lower rate of postoperative hydrocele.
    c. lower rate of varicocele recurrence.
    d. fewer veins ligated.
    e. lower overall complication rate.
5. Which maneuvers should be avoided when bridging a large vasal gap during vasovasostomy?
    a. Mobilization of the vas deferens toward the external inguinal ring
    b. Dissection of the sheath of the convoluted vas deferens off the epididymis and allowing the testis to drop upside-down
    c. Separation of the cauda and corpus epididymis from the testis
    d. Mobilization of the vas deferens toward the internal inguinal ring
    e. Unraveling of the convoluted vas deferens
6. In which of the following scenarios is a testis biopsy least helpful?
    a. Failure to retrieve motile sperm from the epididymis
    b. Sperm retrieval for nonobstructive azoospermia

c. Diagnostic evaluation of men with congenital absence of vas and normal FSH levels
d. Diagnostic evaluation in azoospermic men with normal findings on scrotal examination and normal serum testosterone and FSH levels
e. Sperm retrieval for men diagnosed with Sertoli cell–only pattern in the testes

7. Which of the following scenarios has the lowest rate for sperm return to ejaculate after vasectomy reversal?
   a. Motile sperm in the vas and vasovasostomy
   b. Non-motile sperm in the vas and vasovasostomy
   c. Motile sperm in the vas and unilateral crossed vasovasostomy
   d. Thick, white vasal fluid devoid of sperm and vasovasostomy
   e. Copious clear vasal fluid but no sperm and vasovasostomy
8. In the evaluation for azoospermia, all of the following tests should be considered to confirm a diagnosis of obstructive azoospermia EXCEPT:
   a. transrectal ultrasonography.
   b. testicular biopsy.
   c. serum antisperm antibody assay.
   d. epididymal biopsy.
   e. serum testosterone and FSH assay.
9. Which of the following is TRUE regarding varicocele?
   a. Treatment in infertile men rarely results in improved semen parameters.
   b. Severity of testicular insult is related to the size of the varicoceles.
   c. Severity of testicular insult from varicocele is duration independent.
   d. Because of the severity of testicular insult, repair of large varicoceles is not warranted.
   e. Surgical treatment of subclinical varicoceles results in greater improvement in semen quality than treatment of large varicoceles.
10. After a bilateral vasoepididymostomy, a patient remained azoospermic in two semen analyses until 6 months postoperative,

when the analysis revealed 8 million sperm/mL with 60% motility. What is the next management step?

a. A plan for intrauterine insemination with ejaculated sperm
b. A plan for assisted reproduction with intracytoplasmic sperm injection (ICSI) and ejaculated sperm
c. Cryopreservation of semen
d. No follow-up necessary
e. Scrotal ultrasound

11. Which of the following is a disadvantage of the intussusception vasoepididymostomy?
    a. Inability to assess epididymal fluid for sperm before setting up for anastomosis
    b. Lower patency rate than end-to-side techniques
    c. Difficult hemostasis
    d. Placement of sutures into a collapsed epididymal tubule
    e. Transection of the epididymis required before anastomosis
12. All of the following situations are appropriate for assisted reproduction with ICSI as a first line of treatment EXCEPT:
    a. obstruction with multiple failures of reconstruction.
    b. mild oligoasthenospermia with varicoceles and a female partner of 29 years of age.
    c. Klinefelter syndrome.
    d. only a few viable sperm found in the ejaculate.
    e. post-chemotherapy azoospermia.
13. In which of the following settings are vasovasostomy and vasoepididymostomy contraindicated?
    a. Previous vasectomy more than 20 years ago
    b. Concomitant scrotal pain
    c. Concomitant hydrocele
    d. Concomitant varicoceles
    e. Nonobstructive azoospermia
14. In the presence of epididymal obstruction, which of the following statements is FALSE?
    a. The quality of sperm is better in caput than caudal tubules.
    b. Vasoepididymostomy to the caudal tubules has a better patency rate than to the caput tubules.

c. Vasovasostomy can yield a satisfactory patency rate.
d. A scrotal sonogram may demonstrate epididymal fullness and hydrocele.
e. Intraoperative sperm cryopreservation is possible.

15. Which of the following statements is FALSE regarding microsurgical testicular sperm extraction (TESE)?
    a. Sertoli cell–only pattern is a contraindication for the procedure.
    b. Large seminiferous tubules typically give a higher sperm yield.
    c. It is best performed percutaneously.
    d. It can better preserve the blood supply to testis parenchyma than a non-microsurgical approach.
    e. It should be used in men with nonobstructive azoospermia.
16. All of the following are expected outcomes of varicocele repair EXCEPT:
    a. improved sperm motility.
    b. increased risk of multiple gestation.
    c. improved sperm counts.
    d. improved serum testosterone levels.
    e. return of sperm to the ejaculate in 40% to 50% of azoospermic men.
17. A 30-year-old man presenting with primary infertility was found to be azoospermic on two semen analyses. For which of the following findings is a diagnostic biopsy indicated?
    a. Ejaculate volume below 2 mL with negative fructose
    b. Semen pH less than 7.2
    c. Palpable vasa, normal serum FSH, normal testis volume, and negative antisperm antibodies
    d. Serum FSH of 25 IU/L and 12-mL soft testis
    e. Absence of vasa deferentia and normal serum FSH level
18. In which of the following scenarios is a vasogram indicated at the time of scrotal exploration/possible reconstruction?
    a. Azoospermia, Sertoli cell only on testis biopsy
    b. Azoospermia, testicular volume of 10 mL, FSH value 25 IU/L

c. Azoospermia, normal testicular volume, biopsy revealing active spermatogenesis
d. Azoospermia, no palpable vasa deferential
e. Sperm count 5 million/mL, 5% motility, grade 2 varicoceles bilaterally

19. All of the following diagnoses can be made from a radiocontrast vasogram EXCEPT:
    a. inguinal vasal obstruction.
    b. ejaculatory duct obstruction.
    c. seminal vesicle agenesis.
    d. spermatogenic failure.
    e. partial agenesis of vasa deferentia.
20. All of the following are potential complications of transurethral resection of the ejaculatory ducts EXCEPT:
    a. urinary incontinence.
    b. retrograde ejaculation.
    c. recurrent epididymitis.
    d. testicular atrophy.
    e. contamination of semen with urine.
21. What is the pathogenesis of post-varicocelectomy hydrocele?
    a. Increased testicular venous pressure
    b. Lymphatic obstruction
    c. Soft tissue fibrosis
    d. Arterial injury
    e. Catch-up growth of testes
22. Which of the following is TRUE regarding the ejaculatory duct?
    a. It is a single midline duct formed by the confluence of the seminal vesicle ducts.
    b. It enters into the middle of the verumontanum.
    c. It joins with the prostatic ducts.
    d. It is a paired duct formed by the confluence of each seminal vesicle duct and vasa deferentia.
    e. It enters directly into the vesicle trigone.
23. All of the following are potential complications of vasography EXCEPT:
    a. vasal obstruction at the site of vasography.

b. perivasal hematoma.
c. sperm granuloma at the site of vasography.
d. injury to the vasal artery.
e. retrograde ejaculation.

24. What is the estimated percentage of men who develop antisperm antibodies after vasectomy?
    a. 0% to 20%
    b. 20% to 40%
    c. 40% to 60%
    d. 60% to 80%
    e. 80% to 100%
25. Intraoperatively during a vasectomy reversal, a sperm granuloma is found on the left side. What does this indicate?
    a. Concomitant epididymal obstruction that requires a vasoepididymostomy
    b. Infection requiring postoperative antibiotics
    c. The need for genetic counseling
    d. That sperm will be found at the testicular end of the vas
    e. That the procedure should be abandoned and the patient should undergo re-exploration in 3 months
26. When is the best time to perform vasography?
    a. At the time of diagnostic testis biopsy
    b. At the time of reconstruction, if a prior testis biopsy result was normal
    c. At the time of scrotal ultrasonography with color flow Doppler
    d. At the time of transrectal ultrasonography revealing normal seminal vesicles
    e. At the time of electroejaculation
27. Twelve years after vasectomy, a man was found on routine examination to have asymptomatic sperm granulomas bilaterally. All of the following scenarios are true EXCEPT:
    a. Microrecanalization is possible with the appearance of rare sperm in the ejaculate.
    b. If vasectomy reversal is performed, only bilateral vasovasostomy is likely to be necessary.
    c. The epididymides are unlikely to be indurated.

d. The epididymides are likely to be obstructed.
e. No treatment is necessary for asymptomatic sperm granuloma.

28. A midline cyst compressing the ejaculatory duct is found on a transrectal ultrasonographic scan. What does the presence of sperm in the cyst aspirate suggest?
    a. Congenital absence of vas on at least one side
    b. Nonobstructive azoospermia
    c. Bilateral epididymal obstruction
    d. The possibility of XXY karyotype
    e. Patency of a vas deferens and epididymis on at least one side
29. After transurethral resection of the ejaculatory ducts, the patient develops retrograde ejaculation. What is the next step of management?
    a. Watchful waiting
    b. Intrauterine insemination
    c. ICSI
    d. A trial of pseudoephedrine
    e. Electroejaculation
30. One year after vasovasostomy, a progressive decline in sperm motility and sperm counts is noted. What does this indicate?
    a. Progressive spermatogenic failure
    b. Infection
    c. Arterial injury to the testis and epididymis
    d. Ejaculatory duct obstruction
    e. Stricture of the vasovasostomy
31. In which of the following scenarios would a diagnostic testicular biopsy provide valuable clinical information?
    a. Men with azoospermia, atrophic testes, and an FSH level of 25 IU/L
    b. Men with a 47,XXY karyotype
    c. Men with a fecundity history who seek vasectomy reversal
    d. Men with primary infertility, azoospermia, normal physical examination findings, a normal serum FSH level, and a negative serum antisperm antibody assay
    e. Men with anejaculation caused by high spinal cord injury

32. Which of the following is TRUE regarding retractile testes in adults?
    a. As in the pediatric population, surgical repair is never indicated.
    b. A dartos pouch operation is the treatment of choice.
    c. Simple three-stitch orchiopexy of the tunica albuginea to the dartos, as for torsion prophylaxis, is effective in preventing retraction.
    d. Bilateral orchiopexy is necessary for unilateral retractile testis.
    e. Torsion of the testis is a common complication.
33. Which of the following is TRUE regarding vasoepididymostomy?
    a. End-to-side anastomosis currently has the highest patency rate.
    b. Microsurgical technique does not significantly improve the surgical outcome.
    c. Assisted reproduction with ICSI is a more cost-effective option.
    d. It should be reserved for azoospermia patients with spermatogenic arrest.
    e. It should be performed only on an epididymal tubule containing sperm.
34. When a vasoepididymostomy is performed for fertility reasons, which of the following should be routinely done in the same setting?
    a. Intraoperative epididymal sperm aspiration for sperm cryopreservation
    b. Testicular biopsy for sperm cryopreservation
    c. A touch preparation of testicular tissue
    d. A squash preparation of testicular tissue
    e. A radiocontrast vasogram
35. Which of the following is the most important factor in ensuring a high patency rate after a vasovasostomy?
    a. Age of the patient
    b. Time since vasectomy
    c. Surgeon’s technique and experience
    d. Presence of motile sperm in the vasal fluid

e. Presence of a sperm granuloma at the vasectomy site

36. Which of the following surgical sperm retrieval techniques is inappropriate for the clinical situation indicated?
    a. Percutaneous epididymal sperm aspiration (PESA) for congenital absence of vas
    b. Percutaneous testicular sperm aspiration (PTSA) after failed vasoepididymostomy
    c. Electroejaculation in a man with postretroperitoneal lymph node dissection for left testicular embryonal carcinoma
    d. Microsurgical epididymal sperm aspiration (MESA) for spermatogenic maturation arrest
    e. TESE in a man with azoospermia from chemotherapy

Answers

1. **d. Deferential (vasal) veins. All veins within the cord, with the exception of the vasal veins, are doubly ligated.** Scrotal or gubernacular collateral veins have been demonstrated radiographically to be the cause of 10% of recurrent varicoceles. All external spermatic veins are identified and doubly ligated with hemoclips and divided. The gubernaculum is inspected for the presence of veins exiting from the tunica vaginalis. These are either cauterized or doubly ligated.
2. **b. Hydrocele. The presence of a hydrocele in the presence of excurrent ductal system obstruction is often associated with secondary epididymal obstruction.** Surgeons attempting reconstruction should be aware of the possibility of the need for a vasoepididymostomy.
3. **d. Right: epididymal obstruction and patent right vas above vasectomy site; left: sperm in testicular end of vas and vasectomy site in convoluted vas.** Crossover is indicated in the following circumstances: (1) unilateral inguinal obstruction of the vas deferens associated with an atrophic testis on the contralateral side. A crossover vasovasostomy should be performed to connect a healthy testicle to the contralateral unobstructed vas. (2)

Obstruction or aplasia of the inguinal vas or ejaculatory duct on one side and epididymal obstruction on the contralateral side. It is preferable to perform one anastomosis with a high probability of success (vasovasostomy) than two operations with a much lower chance of success such as unilateral vasoepididymostomy and contralateral transurethral resection of the ejaculatory ducts.

4. **d. Fewer veins ligated.** At the subinguinal level, significantly more veins are encountered, the artery is more often surrounded by a network of tiny veins that must be ligated, and the testicular artery has often divided into two or three branches, making arterial identification and preservation more difficult without using a microscope for the procedure.
5. **e. Unraveling of the convoluted vas deferens.** When large vasal gaps are present, a gauze-wrapped index finger is used to bluntly separate the cord structures from the vas. Blunt finger dissection through the external ring will free the vas to the internal ring if additional abdominal side length is necessary. These maneuvers will leave all the vasal vessels intact. When the vasal gap is extremely large, additional length can be achieved by dissecting the entire convoluted vas free of its attachments to the epididymal tunica, allowing the testis to drop upside-down. If the amount of the vas removed is so large that even these measures fail to allow a tension-free anastomosis, the incision can be extended to the internal inguinal ring, the floor of the inguinal canal cut, and the vas rerouted under the floor, as in a difficult orchiopexy. An additional 4 to 6 cm of length can be obtained by dissecting the epididymis off of the testis from the vasoepididymal junction to the caput epididymis. The superior epididymal vessels are left intact and provide adequate blood supply to the testicular end of the vas. With this combination of maneuvers, gaps up to 10 cm wide can be bridged. The convoluted vas should not be unraveled. This disturbs the blood supply at the anastomotic line.
6. **c. Diagnostic evaluation of men with congenital absence of vas and normal FSH levels. Testis biopsy is indicated in azoospermic men with testes of normal size and consistency, palpable vasa deferentia, normal serum FSH levels, and a**

negative serum antisperm antibody assay. Under these circumstances, biopsy will distinguish obstructive azoospermia from primary seminiferous tubular failure. In the testes of men with congenital absence of vasa, biopsy always reveals normal or at least some spermatogenesis, and biopsy is not necessary before definitive sperm aspiration and in vitro fertilization (IVF) with ICSI.

7. **d. Thick, white vasal fluid devoid of sperm and vasovasostomy.** If the fluid expressed from the vas is found to be thick, white, water insoluble, and like toothpaste in quality, microscopic examination rarely reveals sperm. Under these circumstances, the tunica vaginalis is opened and the epididymis inspected. If clear evidence of obstruction is found (e.g., an epididymal sperm granuloma with dilated tubules above and collapsed tubules below), vasoepididymostomy is performed. When the surgeon is in doubt or is not experienced with vasoepididymostomy, vasovasostomy should be performed. **However, only 15% of men with bilateral absence of sperm in the vasal fluid after barbotage and intensive search will have sperm return to the ejaculate after vasovasostomy.**
8. **d. Epididymal biopsy.** Before surgical reconstruction of the reproductive tract is attempted, spermatogenesis in the patient should be evident. A testicular biopsy may be indicated to confirm the presence of spermatogenesis. Men with a low semen volume should have a transrectal ultrasonographic scan to alert one to the possibility of an additional ejaculatory duct obstruction. For serum and antisperm antibody studies, the presence of serum antisperm antibodies corroborates the diagnosis of obstruction and the presence of active spermatogenesis. At present, this test is of unknown prognostic value and is optional. For serum FSH assay, men with small, soft testes should have serum FSH measured. An elevated FSH level suggests impaired spermatogenesis and potentially a poorer prognosis.
9. **b. Severity of testicular insult is related to the size of the varicoceles. Larger varicoceles appear to cause more damage than small varicoceles; large varicoceles are associated with**

greater preoperative impairment in semen quality than are small varicoceles.

10. **c. Cryopreservation of semen.** With the older end-to-end or end-to-side vasoepididymostomy method, at 14 months after surgery 25% of initially patent anastomoses have shut down. For this reason, we recommend banking sperm both intraoperatively and as soon as sperm appear in the ejaculate postoperatively.
11. **a. Inability to assess epididymal fluid for sperm before setting up for anastomosis.** This method, also known as the *triangulation technique,* was introduced by Berger. There are several advantages of this method versus previous techniques. Two or three sutures placed in the epididymal tubule provide four and six points of fixation, and the anastomosis is virtually bloodless. However, one cannot assess tubular fluid for sperm before the anastomosis setup.
12. **b. Mild oligoasthenospermia with varicoceles and a female partner of 29 years of age.** Assisted reproduction can be offered to men with surgically unreconstructable obstruction such as congenital absence of the vas deferens; men with few viable sperm in the ejaculate; azoospermic men with varicoceles (half of these men will respond to varicocelectomy with return of enough sperm to ejaculate to achieve pregnancy using IVF with ICSI); and men with nonobstructive azoospermia.
13. **e. Nonobstructive azoospermia.** Before surgical reconstruction of the reproductive tract is attempted, spermatogenesis in the patient should be evident. A prior history of natural fertility pre-vasectomy is usually adequate. In other cases, a testicular biopsy may be indicated to confirm the presence of spermatogenesis.
14. **c. Vasovasostomy can yield a satisfactory patency rate.** If clear evidence of obstruction is found, vasoepididymostomy is performed. When there is doubt or the physician is not experienced with vasoepididymostomy, vasovasostomy should be performed. However, only 15% of men with bilateral absence of sperm in the vasal fluid after barbotage and an intensive search will have sperm return to the ejaculate after vasovasostomy.
15. **c. It is best performed percutaneously.** The use of an operating microscope for standard open diagnostic testes biopsy allows

identification of an area in the tunica albuginea free of blood vessels, minimizing the risk of injury to the testicular blood supply and allowing a relatively blood-free biopsy specimen.

16. **b. Increased risk of multiple gestation. Varicocelectomy results in significant improvement in the findings of semen analysis in 60% to 80% of men. Reported pregnancy rates after varicocelectomy vary from 20% to 60%.** A randomized controlled trial of surgery versus no surgery in infertile men with varicoceles revealed a pregnancy rate of 44% at 1 year in the surgery group versus 10% in the control group. In our series of 1500 microsurgical operations, 43% of couples were pregnant at 1 year and 69% at 2 years when couples with female factors were excluded. Microsurgical varicocelectomy results in return of sperm to the ejaculate in 50% of azoospermic men with palpable varicoceles. Repair of large varicoceles results in a significantly greater improvement in semen quality than repair of small varicoceles. In addition, large varicoceles are associated with greater preoperative impairment in semen quality than are small varicoceles, and consequently overall pregnancy rates are similar regardless of varicocele size. Some evidence suggests that the younger the patient is at the time of varicocele repair, the greater the improvement after repair and the more likely the testis is to recover from varicocele-induced injury. **Varicocele recurrence, testicular artery ligation, and postvaricocelectomy hydrocele formation are often associated with poor postoperative results.** In infertile men with low serum testosterone levels, microsurgical varicocelectomy alone results in substantial improvement in serum testosterone levels.
17. **c. Palpable vasa, normal serum FSH, normal testis volume, and negative antisperm antibodies. Men with a positive antisperm antibody assay are always obstructed, and a biopsy is not necessary.** Men with elevated FSH levels and small, soft testes always have nonobstructive azoospermia.
18. **c. Azoospermia, normal testicular volume, biopsy revealing active spermatogenesis.** The absolute indications for vasography are azoospermia, plus complete spermatogenesis with many mature

spermatids on testis biopsy and at least one palpable vas. Relative indications for vasography are severe oligospermia with normal testis biopsy, a high level of sperm-bound antibodies that may be due to obstruction, low semen volume, and poor sperm motility (partial ejaculatory duct obstruction).

19. **d. Spermatogenic failure.** Vasography should answer the questions: Are there sperm in the vasal fluid? Is the vas obstructed? If the testis biopsy reveals many sperm, then the absence of sperm in vasal fluid indicates obstruction proximal to the vasal site examined, most likely an epididymal obstruction. Vasography is done in this case with saline or indigo carmine to confirm the patency of the seminal vesicle end of the vas before vasoepididymostomy. Copious vasal fluid containing many sperm indicates vasal or ejaculatory duct obstruction, and formal contrast vasography is performed to document the exact location of the obstruction. Copious thick, white fluid without sperm in a dilated vas indicates secondary epididymal obstruction in addition to potential vasal or ejaculatory duct obstruction.
20. **d. Testicular atrophy.** Reflux of urine into the ejaculatory ducts, vas, and seminal vesicles occurs after a majority of resections. This can be documented by voiding cystourethrography or by measuring semen creatinine levels. Reflux can lead to acute and chronic epididymitis. Recurrent epididymitis often results in epididymal obstruction. The incidence of epididymitis after transurethral resection is probably underestimated. Symptomatic chemical epididymitis may occur from refluxing urine. If epididymitis is chronic and recurrent, vasectomy or even epididymectomy may be necessary. Even when care has been taken to spare the bladder neck, retrograde ejaculation is common after transurethral resection. Transurethral instrumentation can increase the risk of urethral stricture.
21. **b. Lymphatic obstruction.** Analysis of the protein concentration of hydrocele fluid indicates that hydrocele formation after varicocelectomy is due to lymphatic obstruction.
22. **d. It is a paired duct formed by the confluence of each seminal vesicle duct and vasa deferentia.** The ejaculatory ducts course

between the bladder neck and the verumontanum and exit at the level of and along the lateral aspect of the verumontanum.

23. **e. Retrograde ejaculation. Complications of vasography include stricture, injury to the vasal blood supply, hematoma, and sperm granuloma.** Multiple attempts at percutaneous vasography using sharp needles can result in stricture or obstruction at the vasography site. Careless or crude closure of a vasostomy can also result in stricture and obstruction. Non–water-soluble contrast agents may also result in stricture and should not be employed for vasography. If the vasal blood supply is injured at the site of vasography, vasovasostomy proximal to the vasography site may result in ischemia, necrosis, and obstruction of the intervening segment of vas. A bipolar cautery should be used for meticulous hemostasis at the time of vasostomy to prevent hematoma in the perivasal sheath. Leaky closure of a vasography site may lead to the development of a sperm granuloma, which can result in stricture or obstruction of the vas.
24. **d. 60% to 80%.** Systemic effects of vasectomy have been postulated. Vasectomy disrupts the blood-testis barrier, resulting in detectable levels of serum antisperm antibodies in 60% to 80% of men. Some studies suggest that the antibody titers diminish 2 or more years after vasectomy. Others suggest that these antibody titers persist. However, neither circulating immune complexes nor deposits are increased after vasectomy.
25. **d. That sperm will be found at the testicular end of the vas. A sperm granuloma at the testicular end of the vas suggests that sperm have been leaking at the vasectomy site. This vents the high pressures away from the epididymis and is associated with a better prognosis for restored fertility regardless of the time interval since vasectomy.**
26. **b. At the time of reconstruction, if a prior testis biopsy result was normal.** There is no need to perform vasography at the time of testis biopsy for azoospermia unless immediate reconstruction is planned and the touch or wet-preparation biopsy reveals mature sperm with tails. If performed carelessly, vasography can cause

stricture or even obstruction at the vasography site, which can complicate subsequent reconstruction.

27. **d. The epididymides are likely to be obstructed.** Sperm granulomas form when sperm leak from the testicular end of the vas. Sperm are highly antigenic, and an intense inflammatory reaction occurs when sperm escape outside the reproductive epithelium. Sperm granulomas are rarely symptomatic. The presence or absence of a sperm granuloma at the vasectomy site seems to be of importance in modulating the local effects of chronic obstruction on the male reproductive tract. The sperm granuloma's complex network of epithelialized channels provides an additional absorptive surface that helps vent the high intraluminal pressure in the obstructed excurrent ducts. Numerous animal studies have correlated the presence or absence of sperm granuloma at the vasectomy site with the degree of epididymal and testicular damage. Species that always develop granulomas after vasectomy have minimal damage to the seminiferous tubules. Some studies of men undergoing vasectomy reversal have revealed somewhat higher success rates in men who have a sperm granuloma at the vasectomy site, whereas another large study has not. Although sperm granulomas at the vasectomy site are present microscopically in 10% to 30% of men undergoing reversal, it is likely that, given enough time, virtually all men develop sperm granulomas at the vasectomy site, the epididymis, or the rete testis. When chronic postvasectomy pain is localized to the granuloma, excision and occlusion of the vasa with intraluminal cautery usually relieve the pain and prevent recurrence. Men with postvasectomy congestive epididymitis may be relieved of pain by open-ended vasectomy designed to purposefully produce pressure, relieving sperm granuloma.
28. **e. Patency of a vas deferens and epididymis on at least one side.** The fine-needle aspirate is examined for sperm. If sperm are present, it means at least one vas and epididymis are patent.
29. **d. A trial of pseudoephedrine.** Pseudoephedrine (Sudafed), 120 mg orally, 90 minutes before ejaculation, may prevent retrograde ejaculation. If this is not successful, sperm can be retrieved from

alkalinized urine and used for either intrauterine insemination or IVF with ICSI.

30. **e. Stricture of the vasovasostomy.** Late stricture and obstruction are disappointingly common. **Progressive loss of sperm motility followed by decreasing counts indicates stricture.** A trial of oral prednisone can sometimes prevent progression to azoospermia.
31. **d. Men with primary infertility, azoospermia, normal physical examination findings, and a normal serum FSH level, and a negative serum antisperm antibody assay.** Testis biopsy is indicated in azoospermic men with testis of normal size and consistency, palpable vasa deferentia, and normal serum FSH levels.
32. **b. A dartos pouch operation is the treatment of choice.** When scrotal orchiopexy is performed for retractile testis, a dartos pouch operation should be performed. Simple suture orchiopexy of the tunica albuginea of the testis to the dartos, such as is performed sometimes to prevent torsion, will not prevent retraction of these testes into the groin. Creation of a dartos pouch will keep the testis well down into the scrotum and permanently prevent retraction. This is also the most reliable and safest technique for the prevention of testicular torsion.
33. **e. It should be performed only on an epididymal tubule containing sperm.** Specific treatments for male factor infertility such as microsurgical reconstruction for obstructive azoospermia and varicocelectomy for impaired testes remain the safest and most cost-effective ways of managing infertile men. Microsurgical approaches allow accurate approximation of the vasal mucosa to that of a single epididymal tubule, resulting in marked improvement in the patency and pregnancy rates. If the level of obstruction is not clearly delineated, after the buttonhole opening is made in the tunica, a 70-μm diameter tapered needle from the 10 to 0 nylon microsuture is used to puncture the epididymal tubule, beginning as distal as possible, and fluid is sampled from the puncture site. When sperm are found, the puncture sites are sealed with microbipolar forceps, a new buttonhole is made in the epididymal tunica just proximally, and the tubule is prepared as

described previously. Patency rates with the intussusception technique can exceed 80%. With the classic end-to-side or older end-to-end method, the patency rate is about 70%, and 43% of men with sperm will impregnate their wives after a minimum follow-up of 2 years.

34. **a. Intraoperative epididymal sperm aspiration for sperm cryopreservation.** Once sperm are identified, they are aspirated into glass capillary tubes and flushed into media for cryopreservation.
35. **c. Surgeon's technique and experience.** The responsibilities assumed by the surgeon demand the utmost in judgment and skill. Many of the procedures described in this chapter are among the most technically demanding in all of urology. Acquisition of the skills required to perform them demands intensive laboratory training in microsurgery and a thorough knowledge of the anatomy and physiology of the male reproductive system.
36. **d. Microsurgical epididymal sperm aspiration (MESA) for spermatogenic maturation arrest.** MESA is indicated for men with normal spermatogenesis and unreconstructable obstruction such as congenital bilateral absence of the vas deferens.

Chapter Review

1. If the vas is transected at two different locations, the intervening segment will likely be fibrose because of lack of blood supply.
2. In repairing a hydrocele, the epididymis is often splayed, and one should leave a generous border around the epididymis to avoid injuring it.
3. When sperm are retrieved before any repair, it is prudent to cryopreserve some for future use, if necessary.
4. Following vasoepididymostomy, 50% to 85% of men will have sperm in the ejaculate; about half of these men will foster a pregnancy.
5. The indications for TESE are failure to find sperm in the epididymis and nonobstructive azoospermia.

6. During microsurgical TESE, larger tubules are more likely to yield sperm.
7. About 85% of patients following vasovasostomy will have sperm in their ejaculate; a little more than half will foster a pregnancy.
8. During vasovasostomy, repeated failure to identify sperm in the vasal fluid usually means epididymal obstruction; however, 15% of men with bilateral absence of sperm in the fluid will have sperm return in the ejaculate.
9. Testis biopsy is indicated in azoospermic men with testes of normal size and consistency, palpable vasa deferentia, and normal serum FSH level.
10. Varicocelectomy results in significant improvement in the findings of semen analysis in 60% to 80% of men. Reported pregnancy rates after varicocelectomy vary from 20% to 60%.
11. Men with a positive antisperm antibody assay are always obstructed, and a biopsy is not necessary.
12. A sperm granuloma at the testicular end of the vas predicts presence of sperm in the testicular vasal fluid.
13. Progressive loss of sperm motility followed by decreasing counts indicates stricture.